

Statement of participation

Islam Emad Ahmed Ismail

has completed the free course including any mandatory tests for:

Introducing engineering

This 15-hour free course was an introduction to the broad range of disciplines that come under the heading of engineering, from design to manufacture.

Issue date: 7 October 2025



www.open.edu/openlearn

This statement does not imply the award of credit points nor the conferment of a University Qualification.
This statement confirms that this free course and all mandatory tests were passed by the learner.

Please go to the course on OpenLearn for full details:

<https://www.open.edu/openlearn/science-maths-technology/introducing-engineering/content-section-0>

COURSE CODE: T174_1

Introducing engineering

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Course summary

Engineering encompasses a broad range of disciplines from design to manufacturing. This free course, Introducing engineering, introduces a number of key themes that explore how engineering is undertaken in our modern world. These themes include engineering design, rules that govern engineers, manufacturing for products and electricity generation using solar energy.

Learning outcomes

By completing this course, the learner should be able to:

- understand the characteristics of 'engineering' and the role engineers have played in shaping engineering up to the present and into the future
- understand a range of principles in science, mathematics and engineering in order to make well-founded decisions as part of a design process
- have an appreciation of the design decision-making process when developing new products
- recognise the effects on the conduct of engineering relating to issues such as patents, standards and risk
- understand the use of appropriate candidate/potential materials and processes for the manufacture of a given artefact.

Completed study

The learner has completed the following:

Section 1

Engineering beginnings

Section 2

Engineering by design

Section 3

Engineering to rule

Section 4

Engineering for products

Section 5

Engineering: pushing back the boundaries